A CLOSED SUBSCHEME WHOSE IDEAL IS LOCALLY GENERATED BY ONE ELEMENT, OR BY A SINGLE NONZERODIVISOR, IS LOCALLY PRINCIPAL, RESPECTIVELY AN EFFECTIVE CARTIER DIVISOR

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Statement span 2938-2952 and proof span 2954-2972 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof introduces the finitely many generators of the ideal on the chart and the denominators to be inverted; the author's notation is preserved.

Editorial note. 1 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The localization of a finitely generated ideal at a prime is generated by the localizations of its generators, and a principal generator at the prime can be written with denominators outside the prime. Used at proof lines 2954-2967. Produces a neighbourhood on which the ideal becomes principal, giving the first part.
- An element whose image in a localization at a prime is a nonzerodivisor is a nonzerodivisor on a neighbourhood of the prime, cited from the algebra chapter. Cited at proof lines 2969-2971. Upgrades the pointwise nonzerodivisor condition to a neighbourhood condition, giving the second part.

Source reference. Original lemma 061L: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7765–7765. Same-chapter reference used in the retained author text at source line(s) 2970, 4250, 4869, 5359, 5534, 5635. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let X be a locally Noetherian scheme. Let $D \subset X$ be a closed subscheme corresponding to the quasi-coherent ideal sheaf $\mathcal{I} \subset \mathcal{O}_X$.*

(1) *If for every $x \in D$ the ideal $\mathcal{I}_x \subset \mathcal{O}_{X,x}$ can be generated by one element, then D is locally principal.*
(2) *If for every $x \in D$ the ideal $\mathcal{I}_x \subset \mathcal{O}_{X,x}$ can be generated by a single nonzerodivisor, then D is an effective Cartier divisor.*

Proof. Let $\mathrm{Spec}(A)$ be an affine neighbourhood of a point $x \in D$. Let $\mathfrak{p} \subset A$ be the prime corresponding to x. Let $I \subset A$ be the ideal defining the trace of D on $\mathrm{Spec}(A)$. Since A is Noetherian (as X is locally Noetherian) the ideal I is generated by finitely many elements, say $I = (f_1, \ldots, f_r)$. Under the assumption of (1) we have $I_\mathfrak{p} = (f)$ for some $f \in A_\mathfrak{p}$. Then $f_i = g_i f$ for some $g_i \in A_\mathfrak{p}$. Write $g_i = a_i/h_i$ and $f = f'/h$ for some $a_i, h_i, f', h \in A$, $h_i, h \notin \mathfrak{p}$. Then $I_{h_1 \ldots h_r h} \subset A_{h_1 \ldots h_r h}$ is principal, because it is generated by f'. This proves (1). For (2) we may assume $I = (f)$. The assumption implies that the image of f in $A_\mathfrak{p}$ is a nonzerodivisor. Then f is a nonzerodivisor on a neighbourhood of x by Algebra, Lemma 061L. This proves (2). $\square$